

Education

- 04/2023 - 03/2026** **PhD (Dr. rer. nat)** (Technical University Berlin. Metric Gradient Flows in Measure Spaces: Kernelization and Acceleration. Supervised by Prof. Gabriele Steidl. Referees: Prof. Anna Korba and Prof. José Antonio Carrillo de la Plata. Summa cum laude.)
- 10/2025–04/2026** **BMS Phase II Student** (Graduate school of the three Berlin Universities.)
- 04/2021 – 03/2023** **Mathematics Master's** (Technical University Berlin. Focus on further Functional Analysis topics as well as Topology, Differential Geometry, Complex Analysis, and Statistics. Master's thesis: Wasserstein gradient flows - with an eye towards positive matrix-valued measures. Supervised by Prof. Gabriele Steidl and Dr. Robert Beinert.)
- 10/2017 – 03/2021** **Mathematics Bachelor's** (Technical University Berlin. Focus on Functional Analysis and Differential Equations with a minor in Machine Learning. Bachelor's thesis: Atomic Norm Minimisation for Superresolution. Supervised by Prof. Gabriele Steidl and Dr. Robert Beinert.)

Positions

At the TUM School of Computation, Information and Technology within the Department of Mathematics at Technical University of Munich.

- 05/2026 –** **Postdoc** (supervised by Prof. Massimo Fornasier, Chair of Applied Numerical Analysis, Research Group Data Science and Munich Center for Machine Learning, within the ERC advanced grant NEITALG - Nonlinear Evolutions and Iterative Algorithms: Optimization and Control.)
- 05/2026 –** **Junior Member** (Munich Center for Machine Learning)
- 11/2025** **One-week research visit (Joaquín Míguez at the University Carlos III Madrid)** (Generative modelling with Bernstein approximations of f -divergences.)
- 08/2024 – 09/2024** **One-month research visit (Wuchen Li at the University of South Carolina)** (Accelerated Stein metric gradient flows with general bilinear kernels on Gaussian families.)

In the Applied Mathematics group within the Institute of Mathematics at Technical University Berlin.

- 04/2023 – 14/2026** **PhD candidate** (Funded by a stipend and by the German Federal Ministry of Education and Research under the project “VI-Screen”. From 01/2024, funded by the TU Berlin, I took on teaching responsibilities)
- 06/2021 – 03/2023** **Student research assistant** (Research on Wasserstein gradient flows, writing lecture notes for “Approximation theory”, rewriting the lecture notes for “Convex Analysis” in the setting of infinite-dimensional spaces, and proofreading manuscripts.)

At the Department of Mathematics, Technical University Berlin.

- 10/2019 – 03/2021** **Tutor** (Giving tutorials and correcting homework for the lectures “Functional Analysis I”, “Differential Equations I” and “Linear Algebra for Engineers”.)

Publications

- 01/2026** **V. Stein, José Manuel de Frutos (equal contribution): Approximating f -Divergences with Rank Statistics** (Proceedings of the 43rd International Conference on Machine Learning (ICML'26))

01/2026	V. Stein, Adwait Datar, Nihat Ay: Well-Posed KL-Regularized Control via Wasserstein and Kalman–Wasserstein KL Divergences (Proceedings of the 43rd International Conference on Machine Learning (ICML'26))
12/2025	J. Bresch, V. Stein: Interpolating between Optimal Transport and KL regularized Optimal Transport using Rényi Divergences (Results in Mathematics 81, 23 (2026))
07/2025	R. Duong, V. Stein, R. Beinert, J. Hertrich, G. Steidl: Wasserstein Gradient Flows of MMD Functionals with Distance Kernel and Cauchy Problems on Quantile Functions (ESAIM: Control, Optimisation and Calculus of Variations.)
06/2025	V. Stein, W. Li: Accelerated Stein Variational Gradient Flow (In: Nielsen, F., Barbaresco, F. (eds) Geometric Science of Information. GSI 2025. Lecture Notes in Computer Science, vol 16034. Springer, Cham, pp. 80–90.)
06/2025	R. Duong, N. Rux, V. Stein, G. Steidl: Wasserstein Gradient Flows of MMD Functionals with Distance Kernels under Sobolev Regularization (Philosophical Transactions of the Royal Society A, vol. 383, issue 0243 "Partial differential equations in data science".)
04/2025	V. Stein, S. Neumayer, N. Rux, G. Steidl: Wasserstein Gradient Flows for Moreau Envelopes of f-Divergences in Reproducing Kernel Hilbert Spaces (Analysis and Applications Vol. 24, No. 01, pp. 21-65 (2026).)

Preprints

03/2026	SympFormer: Accelerated attention blocks via Inertial Dynamics on Density Manifolds (with Wuchen Li, University of South Carolina, and Gabriele Steidl, Technical University Berlin.)
09/2025	Towards understanding Accelerated Stein Variational Gradient Flow - Analysis of Generalized Bilinear Kernels for Gaussian target distributions (with Wuchen Li, University of South Carolina.)

Talks

11/2026	SympFormer: Accelerated attention blocks via Inertial Dynamics on Density Manifolds (SIAM Conference on Mathematics of Data Science in Utah, USA.)
11/2026	tbd (SIAM Conference on Imaging Science in Utah, USA.)
09/2026	tbd (BMS – BGSMATH Junior Meeting 2026 in Barcelona, Spain.)
07/2026	Accelerated Stein Variational Gradient Flow (Minisymposium on "New trends in geometric machine and deep learning" at CMAM 2026: Computational Methods in Applied Mathematics in Vienna, Austria.)
06/2026	SympFormer: Accelerated attention blocks via Inertial Dynamics on Density Manifolds (Learning and Optimization in Luminy 2026)
03/2026	SympFormer: Accelerated attention blocks via Inertial Dynamics on Density Manifolds (MFO Workshop 2613 - Flows on Measure Spaces and Applications in Machine Learning in Oberwolfach, Germany.)
02/2026	What does Adam have to do with symplectic manifolds? The geometry behind momentum methods in machine learning (14th BMS Student Conference, Berlin.)
11/2025	Accelerated Stein Variational Gradient Flow (At the colloquium of the Signal Processing Department of University Carlos III Madrid.)

10/2025	Accelerated Stein Variational Gradient Flow (GSI'25: Geometric Structures of Statistical & Quantum Physics, Information Geometry, and Machine Learning (Saint-Malo, France), fully funded by the DAAD.)
09/2025	Wasserstein Gradient Flows for Moreau Envelopes of f-Divergences in Reproducing Kernel Hilbert Spaces (MML'25: Conference on Mathematics of Machine Learning 2025 in Hamburg, Germany.)
04/2025	Accelerated Stein Variational Gradient Flow (Stan Osher's UCLA level set seminar (online).)
09/2024	Interpolating between Optimal Transport and KL regularized Optimal Transport using Rényi Divergences. (University of South Carolina Mathematics Graduate Colloquium)
08/2024	Wasserstein Gradient Flows of MMD Functionals with Distance Kernel and Cauchy Problems on Quantile Functions. (Joint Applied and Computational Mathematics (Changhui Tan & Siming He) and RTG data science seminar (Wuchen Li), University of South Carolina.)
08/2024	Wasserstein Gradient Flows for Moreau Envelopes of f-Divergences in Reproducing Kernel Hilbert Spaces (Stan Osher's UCLA level set seminar (online).)

Poster Presentations at Conferences

07/2026	Foundations of Computational Mathematics 2026 (Wasserstein Gradient Flows for Moreau Envelopes of f -Divergences in Reproducing Kernel Hilbert Spaces (Computational Optimal Transport Session), Accelerated Stein Variational Gradient Flow (Optimization session))
03/2026	TUM's research retreat "Advanced Mathematical Methods in Data Analysis and Signal Processing" in Tutzing, Germany. (Wasserstein Gradient Flows for Moreau Envelopes of f -Divergences in Reproducing Kernel Hilbert Spaces (best poster prize))
02/2026	Gradient structures workshop in Augsburg, Germany. (Wasserstein Gradient Flows for Moreau Envelopes of f -Divergences in Reproducing Kernel Hilbert Spaces)
08/2025	Mathematical and Scientific Machine Learning in Naples, Italy. (Accelerated Stein Variational Gradient Flow)
10/2024	SIGMA (Signal - Image - Geometry - Modelling - Approximation) Workshop at the CIRM in France. (Wasserstein Gradient Flows for Moreau Envelopes of f -Divergences in Reproducing Kernel Hilbert Spaces)
06/2024	LOL: Learning and Optimization in Luminy at the CIRM, France. (Wasserstein Gradient Flows for Moreau Envelopes of f -Divergences in Reproducing Kernel Hilbert Spaces)
03/2024	Workshop on Optimal transport from theory to applications - Interfacing dynamical systems, optimization and machine learning in Berlin, Germany. (Wasserstein Gradient Flows for Moreau Envelopes of f -Divergences in Reproducing Kernel Hilbert Spaces)

Teaching at TU Berlin

Winter 2025/26	Harmonic Analysis (Lecture assistant. Elective BMS advanced module in the Mathematics program.)
Winter 2025/26	Numerical Mathematics I (Tutor. Third-semester compulsory module (in German) in the Mathematics Bachelor's program.)
Summer 2025	Mathematical Physics II - Statistical Mechanics (Lecture assistant. Advanced BMS module for Master's students in the Mathematics program.)

Summer 2025	Probability Theory I (Tutor. Compulsory fourth-semester undergraduate course in the Mathematics program.)
Winter 2024/25	Analysis II for Mathematicians (Tutor. Compulsory module in the Mathematics program, covering multidimensional differentiation.)
Winter 2024/25	Harmonic Analysis (Lecture assistant)
Summer 2024	Convex Analysis (Lecture assistant. Elective advanced module.)
01/2024 - 02/2024	Numerical Mathematics I (Lecture assistant)

Supervised Thesis

03/2025	Roxane Leitheiser, Technical University Berlin (Wasserstein Gradient Flows of the MMD with Riesz Kernels on the Real Line. Bachelor's thesis. First supervisor: Gabriele Steidl.)
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Service

I have reviewed for the Journal of Optimization Theory and Applications (JOTA), Inverse Problems and Imaging (IPI), Transactions on Machine Learning Research (TMLR), the Bayesian Decision-making and Uncertainty Workshop at NeurIPS 2024, and the IEEE/CAA Journal of Automatica Sinica.

I was the reporter for the Oberwolfach Report 26/13 for the workshop Flows on Measure Spaces and Applications in Machine Learning

Awards

Best Bachelor's Thesis Talk at the 17th annual Dies Mathematicus (TU Berlin, 2022).

Best poster prize at the TUM's Research retreat "Advanced Mathematical Methods in Data Analysis and Signal Processing" I received the

IT Skills

I have strong knowledge of Python (including PyTorch) and experience using HPC clusters. Furthermore, I am comfortable with MATLAB and well versed in $\text{\LaTeX}$.

Volunteer Work

In the school year 2022/23, I have been tutoring around 14 seventh-graders in weekly sessions, discussing mathematical puzzles and questions from the German Mathematical Olympiad. I have also served as corrector at the team competition at the Tag der Mathematik 2022 (Mathematics Day) organized by the three Berlin universities, where sixty-nine teams of high schoolers participated.

Language Skills

German: native. English: fluent (spoken and written). French: A2.

References

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